Reg. No. [ ] [18AU55]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

AUTOMOTIVE MATERIALS & COMPONENT MANUFACTURING

Time:3 Hours **Maximum Marks:70**

Instructions:

i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.

ii. Missing data may be suitably assumed

1.	a.	Explain the different conventional materials used in automotive industries.	07
	b.	Explain the aluminum materials used in automobiles.	07
2.	a.	Describe the various types of sensors with its applications.	07
	b.	Explain smart gels with applications.	07
3.	a.	Briefly explain the forging of connecting rod with neat sketch.	09
	b.	Describe the process of stretch forming auto body panels	05
		OR	
4.	a.	Describe the extrusion of valve tappets with a neat diagram.	07
	b.	Explain the process hydro forming of manifolds.	07
5.	a.	Differentiate between gear shaving and gear finishing	08
	b.	Explain the different methods of gear manufacturing.	06
		OR	
6.	a.	Explain gear honing and shaping cutters.	08
	b.	Differentiate between gear hobbing and gear shaping.	06
7.	a.	Illustrate the squeeze casting of pistons with diagrams.	07
	b.	Differentiate between hardening and coating.	07
		OR	
8.	a.	Explain in detail the production of aluminium MMC liners for engine blocks.	08
	b.	Analyze recent developments of auto body panel forming.	06